

Equity Risk Modelling Report

Company Assigned: Oracle Corporation (Ticker: ORCL)

1. Company Profile

Oracle Corporation is a global leader in enterprise software, cloud infrastructure, and database management systems. Founded in 1977 and headquartered in Austin, Texas, the company provides integrated cloud applications and platform services that enable businesses to manage data, analytics, and operations efficiently. In recent years, Oracle has shifted much of its business model toward cloud-based subscription services to compete with Amazon Web Services and Microsoft Azure.

2. Stylised Facts and Frequency Analysis

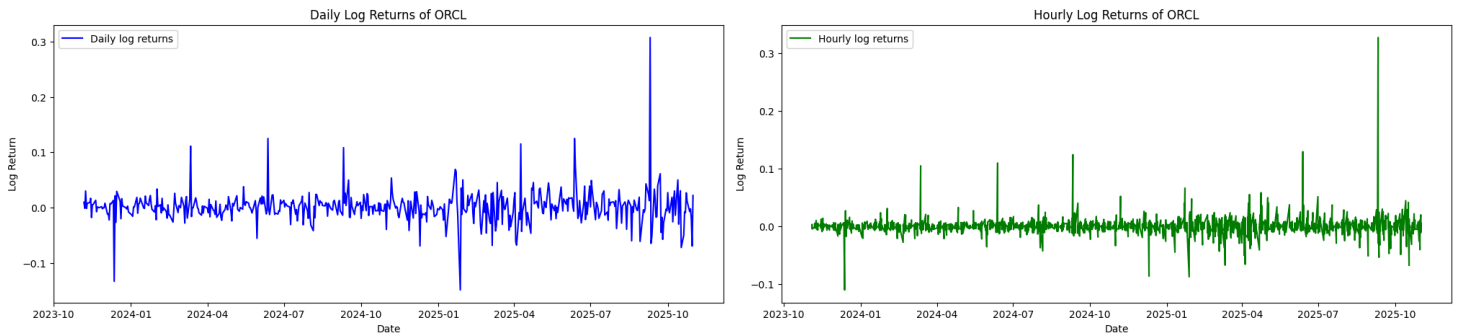


Figure 1: Daily and Hourly returns of Oracle Stock

Both Oracle's daily and hourly return series fluctuate around a mean close to zero, indicating no clear trend over time. The hourly series shows denser, more frequent small movements, reflecting higher trading activity and short-term noise. Several periods in both series exhibit volatility clustering, where large fluctuations are followed by further large movements in both directions. Notably, an extreme outlier around September 2025 is potentially related to firm or market news, which appears to be more pronounced in the hourly series.

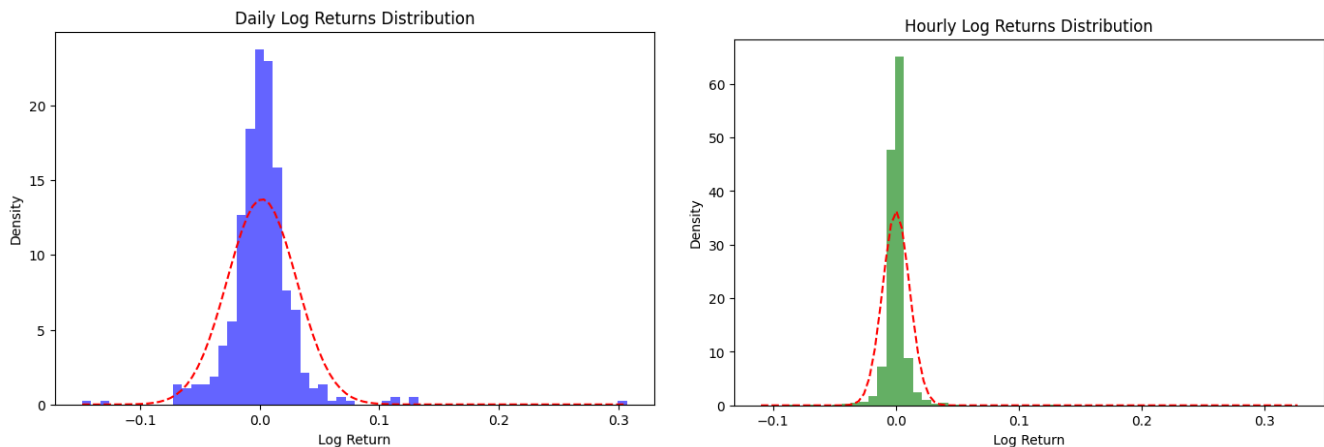


Figure 2: Distribution of Daily and Hourly Returns

Figure 2 displays the daily and hourly log return distributions, each overlaid with a fitted normal distribution curve (red dashed line). Both are sharply peaked around zero with fat tails, indicating a leptokurtosis distribution. This suggests that extreme returns occur more frequently than normal.

The hourly returns are more tightly concentrated near zero and more peaked, while the daily returns exhibit a wider spread with heavier tails, implying greater variability across trading days. A slight left skew in both series also reflects occasional large negative shocks..

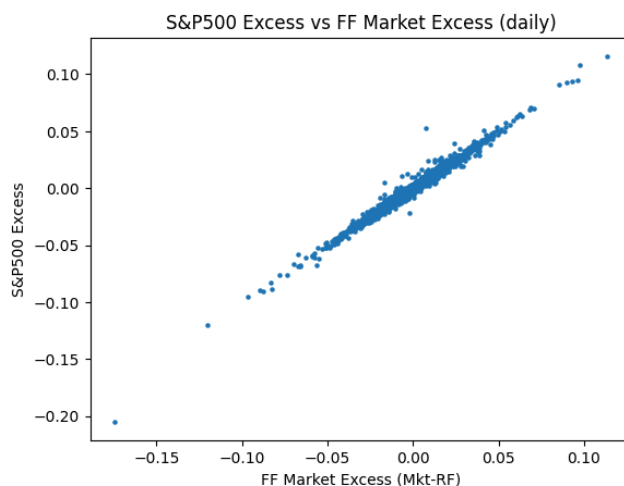
	Daily	Hourly
Mean	0.0018	0.0003
Std Dev	0.0291	0.0110
Skewness	2.2869	8.2985
Kurtosis	27.7626	246.5966
Min	-0.1484	-0.1098
Max	0.3071	0.3272

Figure 3: Descriptive Statistics of Oracle Stock Return

The descriptive statistics confirm the visual findings mentioned above. Both series are non-normally distributed, with high kurtosis values indicating the presence of extreme outliers and heavy tails. The hourly data's exceptionally high kurtosis suggests several large intraday shocks within a stable period, which is a common feature of high-frequency financial data. Overall, Oracle's returns at both frequencies exhibit mean reversion, volatility clustering, fat tails, and non-normality, consistent with the stylised facts of financial return series.

3. Data Matching and Verification

(a) S&P500 & Fama–French Market Excess Returns

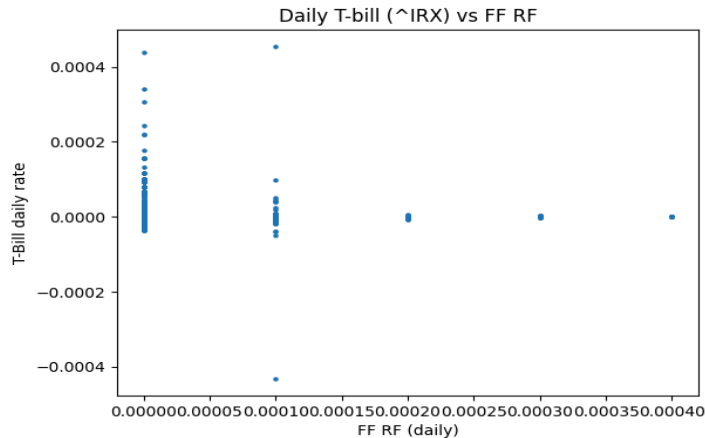


	S&P Excess	Market Excess
S&P Excess	1.0000	0.9905
Market Excess	0.9905	1.0000

Figure 4: S&P 500 Excess Returns and Fama–French Market Excess Returns

The scatter plot shows a strong and clear positive linear relationship for both returns. The close collection of data points along the 45-degree line shows that both variables are good at measuring the same underlying market risk premium. There are a few outliers during times of financial instability or big macroeconomic news, but the overall trend is very stable. This visual pattern is in line with the calculated correlation coefficient of 0.99, which means that the two measures of market excess returns are almost perfectly correlated.

(b) Term T-Bill & Fama–French Risk-Free Rate (RF)



	T-Bill	RF
T-Bill	1.0000	-0.0678
RF	-0.0677	1.0000

Figure 5: Term T-Bill and Fama–French Risk-Free Rate (RF)

The scatter plot that compares the daily T-Bill rate to the Fama–French risk-free rate (RF) shows a weak and scattered relationship, with a correlation coefficient of -0.0677 . Both are short-term, risk-free benchmarks, but they differ mostly because of how the data is put together and reported.

The T-Bill rate is shown as an annualised percentage yield, while the Fama–French RF is shown as a daily rate. Even after adjusting for the effects of compounding, the two measures align only approximately. The dispersed pattern in the plot likely arises from minor differences in timing, data frequency, and scaling conventions. This highlights the inherent difficulty of reconciling financial data sourced from distinct methodologies and reporting standards.

4. Capital Asset Pricing Model (CAPM)

(a) CAPM & β Estimation

R-squared	0.265
Adjusted R-squared	0.264

Coefficient	Estimates	t-stat	p-value
α (Intercept)	0.0007	2.751	0.006
β (Market)	1.3058	59.576	0.000

Figure 6: CAPM Regression

Oracle's stock shows strong sensitivity to market movements, with a beta of 1.31, meaning a 1% rise in the market's excess return raises Oracle's by about 1.31%. The small but significant positive alpha suggests a modest abnormal return beyond what market risk explains, indicating that CAPM only partially captures Oracle's return dynamics.

Positive alpha suggests that some abnormal returns cannot be fully explained by the market factor alone. Short-term inefficiencies, missed risk factors, or the impact of high-frequency daily data could all be the cause of this deviation.

However, the R-squared value of 0.265 and the strong t-statistic for beta indicate that the model explains a notable portion of Oracle's return variation. To mitigate the effects of volatility-related noise and improve explanatory power, a Fama–French multifactor model could be used, or alternatively use lower-frequency data for more reliable estimation.

(b) Replication Portfolio with S&P500 & T-Bill

R-squared	0.253
Adjusted R-squared	0.253

Coefficient	Estimates	t-stat	p-value
α (Intercept)	0.0008	3.155	0.002
β (Market)	1.2608	57.949	0.000

Figure 7: Replication Portfolio Regression

A beta greater than one suggests that Oracle's stock tends to amplify market trends as it rises faster during market upswings and declines more sharply in downturns. The positive and significant alpha indicates a small abnormal return that market risk alone cannot explain. This suggests that while the CAPM generally holds for Oracle, it does not fully capture all factors influencing its returns. Based on these results, a replication portfolio was created by allocating 126.1% to the S&P 500 and financing -26.1% through T-Bills.

The cumulative excess return graph in the appendix indicates that this replication portfolio closely mirrors Oracle's overall performance but delivers greater long-term growth because of its leveraged exposure. This outcome reinforces that the CAPM accounts for most of Oracle's return behaviour, while residual differences are likely driven by firm-specific factors or unmodelled risks.

(c) Historical expected return, variance & Sharpe ratio for Oracle and its replication portfolio

	Oracle (Excess Return)	Replication Portfolio
Mean	0.2918	0.2918
Variance	0.2134	0.0541
Standard Deviation	0.4620	0.2325
Sharpe Ratio	0.6317	1.2552

Figure 8: Oracle Vs Replication Portfolio Excess Return

Oracle and its replication portfolio have an annualised expected excess return of about 29.2%. On the other hand, replication portfolio shows a substantially lower annualised variance than Oracle, suggesting that it generates comparable returns with a lot less volatility. As a result, Oracle's Sharpe ratio is almost twice that of the replication portfolio which indicates that it provides a higher risk-adjusted return.

Under the CAPM framework, this outcome suggests that an investor could more efficiently replicate Oracle's returns by leveraging the market portfolio and borrowing at the risk-free rate. Thus, the replication portfolio achieves a better return per unit of risk which makes it a more appealing investment for a rational and risk-averse individual. Oracle's lower Sharpe ratio reflects the presence of firm-specific and unique risks that are not compensated by proportionally higher expected returns.

5. APT / Factor Models

(a) Fama-French 5-Factor Model Estimation

R-Squared	0.091
Adj. R-Squared	0.074

Factor	Coefficient	t-stat	p-value
Mkt-RF	-0.0095	-0.235	0.814

AIC	-1250
BIC	-1228

SMB	0.0826	1.497	0.136
HML	0.0723	0.990	0.323
RMW	-0.2690	-3.776	0.000
CMA	-0.0579	-0.548	0.584

Figure 9: Fama-French 5-Factor Model Fit

The Arbitrage Pricing Theory (APT) assumes that assets returns can be explained by their sensitivities to multiple systematic risk factors. In this analysis, we estimate and test an APT-type pricing model using the Fama-French 5-Factor framework, which includes market risk (Mkt-RF), size (SMB), value (HML), profitability (RMW) and investment (CMA) factors as listed above.

The RMW factor is the only statistically significant factor with a negative coefficient (p-value <0.05). This may indicate a tilt toward low-profitability or turnaround characteristics. The remaining factors (Mkt-RF, SMB, HML, and CMA) are all statistically insignificant, indicating weak sensitivity to market, size, value, and investment effects. This suggests that these systematic risks are either not materially priced for this stock, or that its return dynamics lie largely outside the exposures captured by the five-factor framework.

(b) Factor & Factor Loading Interpretation

Based on the results of the Fama-French Five-Factor Model regression, below is an interpretation of the factors and their loadings :

Mkt-RF (Market Risk Premium) represents the excess return of the overall market (proxied by a broad market portfolio) over the risk-free rate, with coefficient (β) indicating stock's sensitivity. Stock tends to move in the direction of the market if $\beta > 0$, stock is more volatile than the market if $\beta > 1$.

SMB (Small Minus Big) captures the historical tendency for small-cap stocks to outperform large-cap stocks, suggesting stock's sensitivity to the size factor. A positive coefficient suggests the stock has small-cap, while a negative coefficient suggests it has a large-cap.

HML (High Minus Low) accounts for the value premium, the tendency for value stocks (high book-to-market ratio) to outperform growth stocks, the coefficient (β) indicates the stock's sensitivity to this value factor. The stock is a value stock if $\beta > 0$, and a growth stock if $\beta < 0$.

RMW (Robust Minus Weak) represents the difference between the returns of companies with robust profitability and those with weak profitability, the coefficient (β) indicates the stock's sensitivity to profitability. Stock associated with higher profitability has $\beta > 0$, while $\beta < 0$ is associated with companies with lower profitability.

CMA (Conservative Minus Aggressive): This factor captures the difference between the returns of companies that invest conservatively and those that invest aggressively. The coefficient (β) on CMA indicates the stock's sensitivity to this investment factor. $\beta > 0$ suggests the stock is associated with companies that invest more conservatively, while $\beta < 0$ suggests it's associated with companies that invest more aggressively.

The factor loadings (the coefficients for Mkt-RF, SMB, HML, RMW, and CMA) represent the estimated sensitivity of Oracle's excess returns to each of these systematic risk factors. In the context of the APT, these loadings, along with the factor risk premiums, determine the expected return of the stock in addition to the market premium. A statistically significant coefficient suggests that the stock has a meaningful exposure to that particular factor beyond its market exposure. The intercept term (const) in the regression represents the abnormal return (alpha) that is not explained

by the included factors.

(c) Momentum Factor Extension

R-Squared	0.095	Factor	Coefficient	t-stat	p-value
Adj. R-Squared	0.075	Mkt-RF	-0.0129	-0.319	0.750
AIC	-1249	SMB	0.0777	1.403	0.162
BIC	-1223	HML	0.0706	0.967	0.334
		RMW	-0.2692	-3.780	0.000
		CMA	-0.0696	-0.656	0.512
		MOM	-0.2196	-1.095	0.274

Figure 10: Fama-French 5-Factor Model Fit after implementation of Momentum

The addition of the MOM (momentum) factor, the tendency of stocks with recent high returns to continue outperforming, on top of the Fama-French Five-Factor Model may represent a priced risk factor. As the Five-Factor Model (Mkt-RF, SMB, HML, RMW, CMA) all observed minor or no meaningful changes statistically with the inclusion of the MOM factor, adding this factor itself which introduces a sixth systematic factor consistent with APT's flexibility in factor selection, is also not statistically significant. Its inclusion does not materially improve the model fit, as R-Squared increases only from 0.091 to 0.095. RMW factor remains the only significant factor, reinforcing the idea that profitability-related risks are priced for their stock.

(d) Oil Price Sensitivity

R-Squared	0.082	Factor	Coefficient	t-stat	p-value
Adj. R-Squared	0.048	Mkt-RF	0.1049	2.312	0.022
AIC	-810.2	SMB	-0.0024	-0.034	0.973
BIC	-788.2	HML	0.0801	1.043	0.299
		RMW	-0.1126	-1.429	0.155
		CMA	-0.0247	-0.238	0.812
		CL=F	0.0519	0.823	0.412

Figure 11: Fama-French 5-Factor Model Fit after addition of oil price changes

The p-value for the CL-F factor is 0.412. Since this p-value is greater than the significance level of 0.05, we cannot reject the null hypothesis that the coefficient for oil price changes is zero. This suggests that relative changes in oil prices are not a statistically significant additional source of risk that pays a premium on Oracle's stock, after accounting for the Fama-French 5-Factors.

This finding is not surprising, as Oracle Corporation is primarily a technology company providing software, database and cloud services. Its core business operations are not directly tied to the production or consumption of oil. Therefore, direct exposure to oil price fluctuations is likely minimal. While there could be indirect effects through the broader economy impacting business

spending, the regression results do not indicate a statistically significant direct or indirect pricing risk related to oil price changes for Oracle within this model.

(e) Model Selection

Factor / Model	Fama-French 5-Factor Model	Fama-French 5-Factor + Momentum Model	Fama-French 5-Factor + Oil Model
Mkt-RF	1.2494	1.2569	1.3455
SMB	1.2684	1.2768	1.2827
HML	2.0011	2.0020	1.9820
RMW	1.4915	1.4915	1.4251
CMA	1.970	1.9906	1.8569
MOM	-	1.0337	-
CL=F	-	-	1.0190

Figure 12: Comparison of VIF Models

In order to select the best model from amongst the three models as shown above, the model itself has to satisfy several conditions under the selection criteria .

First, the preferred model should exhibit low multicollinearity, typically assessed through Variance Inflation Factors (VIF). Values above 5 or 10 are considered problematic, but all three models perform well, with VIFs clustered around 1–2, which are well below concerning levels. Second, the model should include statistically significant factors (p -value < 0.05). Across regressions, only the RMW (profitability) factor consistently meets this threshold, whereas MOM (momentum) and CL=F (oil price changes) remain insignificant for this stock. Third, the model should have a higher adjusted R^2 , reflecting better explanatory power after accounting for the number of predictors. Finally, lower information criteria values (AIC and BIC) indicate a more parsimonious specification that balances fit and complexity.

Among the three models, the baseline Fama–French Five-Factor model offers the best trade-off between parsimony and interpretability, with RMW as the dominant priced risk factor. It achieves the second-highest adjusted R^2 (0.091), only 0.004 below the Momentum Model, indicating that the latter provides minimal additional explanatory power. Furthermore, its AIC (–1250) and BIC (–1228) are the lowest among all models, reinforcing its relative efficiency. Although adding momentum and oil factors is theoretically justified, these extensions don’t enhance model fit or significance. Therefore, the Fama–French 5-Factor model remains the most appropriate APT-type specification for Oracle’s stock, offering a robust framework for explaining its return dynamics.

6. ARMA models and Forecasting

(a) ACF & PACF for Returns and Squared Returns

ACF and PACF display similar patterns across both returns and squared returns, indicating non-stationarity and offering limited insights for model selection. Therefore, ADF and KPSS tests were conducted to confirm stationarity.

	p-value
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	With constant	With constant & trend
ADF	0.000	0.000
KPSS	0.038	0.100

Figure 13: ADF and KPSS Test Results

The ADF test checks for the presence of a unit root, where a significant result indicates stationarity, while the KPSS test assumes stationarity as its null hypothesis and detects deviations from it. At the 95% confidence level, ADF results suggest stationarity, while KPSS indicates non-stationarity under the constant assumption. Hence, the series was first-differenced.

After differencing, both tests confirm stationarity, with the ACF/PACF plots revealing clearer patterns suitable for ARMA model identification.

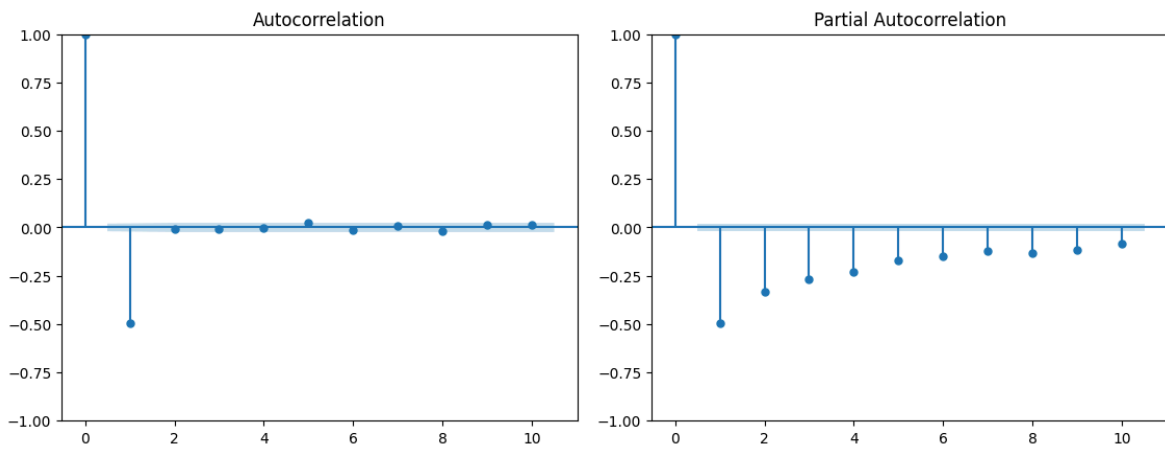


Figure 14: ACF & PACF on Returns

It's evident that only the first lag in the ACF is significant, with subsequent lags fluctuating around zero, suggesting an AR range of 0–1. The PACF in the other hand shows a gradual decay until around lag 5 before stabilising, indicating a broader MA range of 0–4 for model consideration.

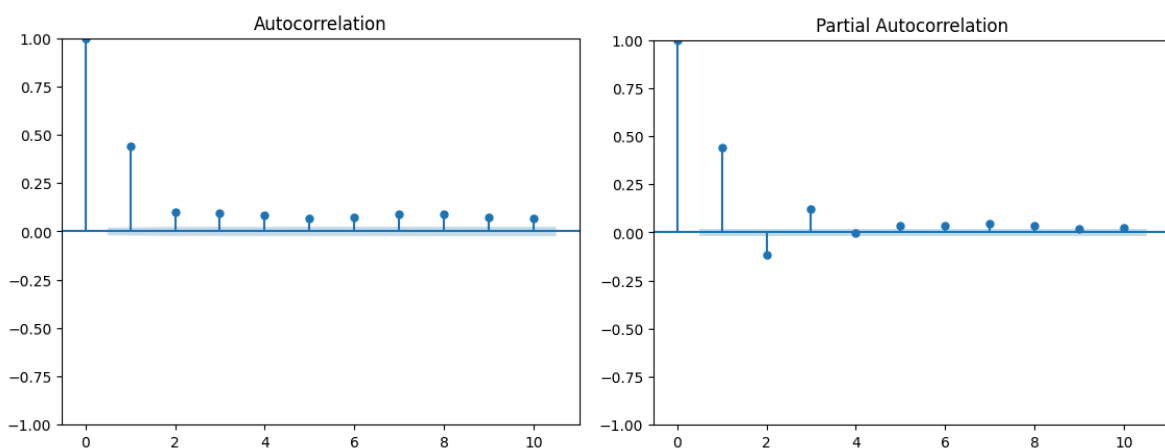


Figure 15: ACF & PACF on Squared Returns

In this ACF, only the first lag stands out, while subsequent lags fluctuate around zero, suggesting an AR range of 0–1 similar to the returns series. Interestingly, this PACF displays alternating fluctuations around the mean, with three significant lags, indicating a reasonable MA range of 0–3.

(b) ARMA Model Selection

Since first differencing was required to achieve stationarity, $d=1$ is applied when estimating ARIMA models for both returns and squared returns.

ARMA Model on Returns			ARMA Model on Squared Returns		
p	q	BIC	p	q	BIC
0	3	-41920.22	0	1	-84528.10
0	1	-41912.60	0	2	-84027.45
0	2	-41911.29	0	2	-74730.36

Figure 16: Comparison of ARMA Models

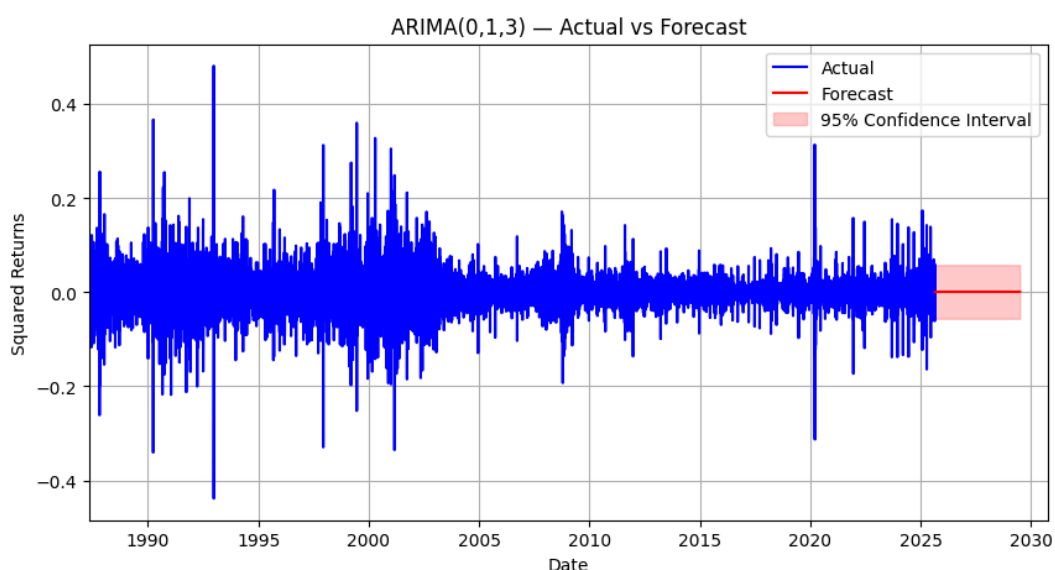
The optimal model for returns is ARIMA(0,1,3), while that for squared returns is ARIMA(0,1,1). Although the squared returns model achieves a substantially lower BIC, it captures only volatility magnitude rather than directional movement. Hence, ARIMA(0,1,3) on returns was selected for subsequent forecasting and residual diagnostics.

From the model summary (Appendix A), the selected ARIMA(0,1,3) model consists of three moving average terms, with MA(1) and MA(3) statistically significant at the 5% level. The Ljung–Box test confirms that the model adequately captures linear dependence, as no significant autocorrelation remains in the residuals (p -value = 0.92 > 0.05), supporting the white noise hypothesis. However, both the Jarque–Bera and heteroskedasticity tests reveal significant p -values, indicating non-normality and non-constant variance patterns. These findings suggest potential non-linear dependence remaining in the residuals.

(c) Non-linear Dependence Residual Analysis

The BDS test examines non-linear dependence in residuals, with the null hypothesis that the data are independent and identically distributed (i.i.d.). Serving as a diagnostic tool, it detects hidden structures not captured by linear models such as ARIMA. Consistent with earlier findings, the BDS test for Oracle's returns yields a significant p -value (<0.05), indicating remaining non-linear dependence in the residuals. This suggests that the linear ARIMA model does not fully capture the underlying dynamics, potentially limiting its forecasting ability.

(d) Analytical & Dynamic Long-run Forecast



Long-run Forecast	Dynamic Forecast	Difference
0.01277%	0.01102%	0.00175

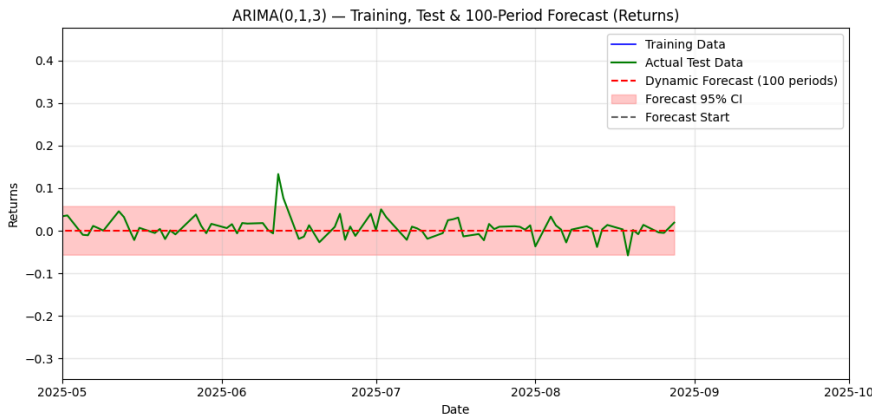
Figure 17: Results of Long-run Forecast

The long run mean forecast, computed from sample mean, differs from the dynamic forecasting by 0.00175%. With no AR terms, the sample mean remains unchanged, thus the long run forecast was computed through the formula below:

$$\mu = \frac{\sum r_i}{n} \quad \{\mu = \text{mean}, \sum r_i = \text{sum of returns}, n = \text{number of returns}\}$$

The dynamic forecast initially showcases returns of 0.0307%, which converges to 0.01102% as MA(3) effects dissipate. The difference demonstrates the model's ability to distinguish shocks from persistent components as recent highs are treated as temporary. The near-zero forecasts align with the efficient market hypothesis showing past data have minimal predictive power.

(e) Out-of-sample Evaluation



Range Dynamic Forecast	Range Actual Data
[0.0008, 0.0054]	[-0.05, 0.13]

Figure 18: Forecast of Out-of-sample Data

The near flat dynamic forecast showcases that the model is unable to capture the volatility of the returns. The wide confidence interval does capture most of the training data however, due to its wideness, it renders the model impractical when it comes to decision making. The poor point forecast, partnered with a wide confidence interval, showcases that the past returns do not have the capability to predict future forecasts.

(f) ARMA & White Noise Model Comparison

ARMA Forecast	Whitenoise	Difference
0.0285	0.0284	-0.0001

Figure 19: RMSE of ARMA Model & White Noise Model

The ARIMA (0,1,3) model has a higher RMSE than the white noise benchmark, with negligible difference of -0.000137. This indicates that the ARIMA model has no predictive capability beyond white noise. It implies that model complexity does not improve forecast accuracy. Overall the findings align with the weak-form efficient market hypothesis that states past price has no capability to improve on forecasts of future returns.

Appendix

A - ARIMA(0,1,3) Model Summary

SARIMAX Results						
Dep. Variable:	ORCL	No. Observations:	9902			
Model:	ARIMA(0, 1, 3)	Log Likelihood	20978.512			
Date:	Sat, 01 Nov 2025	AIC	-41949.024			
Time:	03:56:10	BIC	-41920.223			
Sample:	0	HQIC	-41939.270			
	- 9902					
Covariance Type:	opg					
	coef	std err	z	P> z	[0.025	0.975]
ma.L1	-1.0308	0.008	-126.897	0.000	-1.047	-1.015
ma.L2	-0.0119	0.010	-1.244	0.214	-0.031	0.007
ma.L3	0.0428	0.007	6.437	0.000	0.030	0.056
sigma2	0.0008	6.23e-06	135.492	0.000	0.001	0.001
Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	86535.80			
Prob(Q):	0.92	Prob(JB):	0.00			
Heteroskedasticity (H):	0.23	Skew:	0.43			
Prob(H) (two-sided):	0.00	Kurtosis:	17.46			